Modern Algebra 2

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Abstract

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Contents

Chapter 1

Rings

Lecture 1: Syllabus Day

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Definition 1.1: Ring

Let R be a set together with two binary operations

$$+, \bullet : R \times R \to R$$

such that $\forall a, b, c \in R$,

- $a + b = b + a$;
- $(a + b) + c = a + (b + c)$;
- $\exists 0 \in R$ such that $a + 0 = a$ for all $a \in R$;
- $\forall a \in R$, $\exists b \in R$ such that $a + b = 0$;
- $(ab)c = a(bc)$;
- $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

We then call $(R, +, \cdot)$ a ring.

Note that $(R, +)$ is an abelian group. There then clearly exists a forgetful functor **Ring** $\to$ **Grp** that maps $(R, +, \cdot) \mapsto (R, +)$. Examples of rings include $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under the standard definitions of addition and multiplication. However, these rings are all commutative, and many are in fact fields. There is more to ring theory than commutative rings. Oh also $\mathbb{Z}_n$ is a ring $\forall n \in \mathbb{N}$.

Note that since left multiplication is a faithful group action on itself, every row/column of the cayley table in some group has each number appear exactly once.

We take the convention that rings need not have 1. Homomorphisms thus need not preserve 1, except for epimorphisms. An example of a ring without 1 is $2\mathbb{Z}$. Also note that rings need not be commutative; for example, $M_2(\mathbb{R})$. We generally divide rings into commutative rings and noncommutative rings. the category of commutative rings is denoted **CRing**.

Definition 1.2: Unity

Let R be a ring with some $1 \in R$ such that $\forall r \in R$, $1r = r1 = r$. We say 1 is the unity element of R.

Definition 1.3: Polynomial Ring

Let R be a ring. Define $R[x]$ as the polynomial ring

$$R[x] \coloneqq \left\{ \sum_i a_i x^i \mid a_i \in R \right\}$$

of *finite* linear combinations in the indeterminant x.

Lecture 2: Basic Ring Properties

Fri 24 Jan 2025 13:27

Let R be a ring without unity. Define $S = R \times \mathbb{Z}$ or $S = r \times \mathbb{Z}_{\text{char}(R)}$. Define multiplication by

$$(r, n)(s, m) = (rs + ns + mr, nm).$$

Then $(0, 1)$ is a unity and there exists an inclusion $R \hookrightarrow S$ which is also a ring homomorphism by the map $r \mapsto (r, 0)$.

The quaternions $\mathbb{H}$ are defined as the set of linear combinations of the set $\{1, i, j, k\}$ wherein multiplication satisfies $ijk = -1$, $i^2 = -1$, $j^2 = -1$, and $k^2 = -1$. They are intended as an extension of the complex numbers $\mathbb{C}$, in a similar way to how the complex numbers extend $\mathbb{R}$. However, this definition of multiplication necessitates $ij = -ji$, meaning $\mathbb{H}$ is a noncommuative ring. Furthermore, it can be shown (with some difficulty) that all elements of $\mathbb{H}$ have multiplicative inverses. So $\mathbb{H}$ is a noncommutative ring with unity with multiplicative inverses. We call this a divison ring.

Definition 1.4: Division Ring

Let R be a ring with unity, such that $\forall r \in R$, $\exists r^{-1} \in R$ such that $rr^{-1} = r^{-1}r = 1_R$. Then R is a division ring.

Going back down in dimension to $\mathbb{C}$ gives more structue. Elements in $\mathbb{C}$ do, in fact, commute under multiplication; this makes $\mathbb{C}$ into a field.

Definition 1.5: Field

Let k be a commutative ring with unity, such that for all $r \in k$, there exists $r^{-1} \in k$ such that $rr^{-1} = 1_k$. Then k is called a field.

Definition 1.6: Direct Product

Let R, S be rings. Define the direct product $R \times S$ as the cartesian product of the sets R, S together with coordinate-wise addition and multiplication. Moreover, for any set of rings $R_1, \ldots, R_n$, define the product $R_1 \times \cdots \times R_n$, which is defined in the same way.

The direct product satisfies the universal property for products in **Ring**, but there are more multiplicative structures that can be endowed to the underlying abelian group $R_1 \oplus \cdots \oplus R_n$, which we may or may not see when we do polynomial rings. We may write $\oplus$ instead of $\times$ in some commutative cases.

Proposition 1.1

Let R be a ring, and let $a, b, c \in R$. Then

- $a0 = 0a = 0$;
- $a(-b) = (-a)b = -ab$;
- $(-a)(-b) = ab$;
- $a(b-c) = ab - ac$ and $(a-b)c = ac - bc$;
- $(-1)a = -a$;
- $(-1)(-1) = 1$.

Proof. (1) Many of these are trivial, so I will only prove a few. Let $a \in R$. Then

$$0a = (0+0)a = 0a + 0a \implies 0a - 0a = 0a - 0a - 0a \implies 0 = 0a.$$

(3) Now let $a, b \in R$ and note that $(-a+a)(-b) = 0(-b) = 0$, and thus

$$(-a)(-b) + a(-b) = 0.$$

By property 2,

$$(-a)(-b) + a(-b) = (-a)(-b) - (ab).$$

Therefore, $(-a)(-b) = ab$.

Statement holds for $n = 1, 2$ trivially. Let n even and suppose the statement holds $\forall m \leq n$. We have

$$x^{n+2} = \left(x^{\frac{n}{2}+1}\right)^2 = 0 \iff x^{\frac{n}{2}+1} = 0.$$

Therefore, $x^{n+2} = 0$ iff $x = 0$. Now we have immediately

$$x\left(x^{n+1}\right) = x^{n+2}.$$

If $x \neq 0$ but $x^{n+1} = 0$, then we have $x^{n+2} = 0$ for $x \neq 0$, contradiction. ■

Lecture 3: Rings with Unity

Mon 27 Jan 2025 13:25

Since every ring can be "given" unity, we generally assume rings have unity, because it makes life much easier.

Definition 1.7: Unit

Let R be a ring with unity, and let $a \in R$ such that there exists $b \in R$ such that $ab = 1$. Then a, b are units, and we say b is the multiplicative inverse of a.

Proposition 1.2

Let R be a ring with unity. Then

- the unity is unique;
- If an element $a \in R$ has a multiplicative inverse, then this inverse is unique.

Proof. Let $1_R, 1_S$ be unities of R. Then $1_R 1_S = 1_R = 1_S$ since they are both unities. Let $a \in R$ be a unit, and let b, c be inverses. Then $1 = ab = ac$, and it follows that

$$bab = bac \implies 1b = 1c \implies b = c.$$

■

Proposition 1.3

Let R be a ring with unity. Then the set $U(R)$ of units of the ring R is a group with respect to ring multiplication.

Proof. Note that $1 \in U(R)$, so clearly $U(R)$ has an identity. We also have that all elements have inverses by definition, and associativity follows from the ring axioms. Let $a, b \in U(R)$. Since $(ab)^{-1} = b^{-1}a^{-1}$, and $a^{-1}, b^{-1} \in U(R)$, we have that $ab \in U(R)$, and $U(R)$ is a group. ■

Theorem 1.1

$$U(\mathbb{Z}_n) = \{r \in \mathbb{Z}_n \mid \gcd(r, n) = 1\}.$$

Proof. If the gcd is not 1, then there exists some multiple $mr = 0$ of r with $m < n$. If r is a unit, then $mrr^{-1} = 0$, so $m = 0$, so $\gcd(r, n) = 1$, which is a contradiction. ■

Cancellation:

Proposition 1.4

Let R be a unital ring and let $a \in R$ be a unit. Then $ab = ac$ implies $b = c$.

Proof. Let $ab = ac$. Then $a^{-1}ab = a^{-1}ac$, and thus $b = c$. ■

An example of a finite field is $\mathbb{Z}_p$ for p prime.

Definition 1.8: Subring

Let R be a ring and let $S \subseteq R$ be a subset. Then S is a subring if it is a ring given the same operations as R; S is a subring if the inclusion map $S \hookrightarrow R$ is a ring homomorphism.

Proposition 1.5

Let $S \subseteq R$, such that for all $a, b \in S$, $a - b \in S$ and $ab \in S$. Then S is a subring of R.

Proof. Trivial. ■

Lecture 4: Domains

Mon 03 Feb 2025 13:22

Observe that

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

Why not apply this notion of a fraction to domains other than $\mathbb{Z}$?

Proposition 1.6

Let D be a domain. Then there exists a field F called the field of fractions $\text{Frac}(D)$, containing (the inclusion of) D as a subring.

Proof. The construction is as follows. Consider the collection

$$S = \{(a, b) \in D \times D \mid b \neq 0\}.$$

Define an equivalence relation $\equiv$ on S as

$$(a, b) \equiv (c, d) \iff ad = bc.$$

Define F as the set of equivalence classes $D/\equiv$. We can define

$$\frac{a}{b} \coloneqq [(a, b)].$$

The operations $+, \cdot$ are defined as

$$[(a, b)] \cdot [(c, d)] = [(ac, bd)], \qquad [(a, b)] + [(c, d)] = [(ad + bc, bd)].$$

It remains to be shown F is a field. Showing the operations are well-defined is messy and I don't care. Showing everything else is even messier and I care even less. By the method of apathy, F is a field. Finally, note that we can define the inclusion $D \hookrightarrow F$ as $d \mapsto d/1$, and this map is in fact an isomorphism. ■

$\text{Frac}(D)$ satisfies the universal property that it is initial with respect to the property of containing D as a subring. That is, any field containing D must contain $\text{Frac}(D)$.

Definition 1.9: Polynomial Ring

Let R be a commutative ring. Define the polynomial ring $R[x]$ as the set of R-linear combinations in the indeterminate x:

$$R[x] \coloneqq \left\{ \sum_{i=0}^{n} a_i x^i \mid \{x_i\}_{i=0}^{k} \subseteq R \right\}.$$

Equality is defined componentwise and addition and multiplication are defined in the usual way.

Definition 1.10: Degree

Let R be a commutative ring, and let $f(x) \in R[x]$. Then $\deg f$ is the greatest n such that $a_n \neq 0$, where

$$f(x) = \sum_{i=0}^{k} a_i x^i.$$

The degree of the 0 polynomial is $-\infty$ because woooo

Proposition 1.7

If R is an integral domain, then $R[x]$ is an integral domain.

Lecture 5: Polynomial Rings

Wed 05 Feb 2025 13:27

Definition 1.11: Ring Homomorphism

Let $(R, +_R, \bullet_R)$ and $(S, +_S, \bullet_S)$ be rings. A function

$$\phi : R \to S$$

is called a *ring homomorphism* if and only if

$$\phi(a +_R b) = \phi(a) +_S \phi(b), \qquad \phi(a \bullet_R b) = \phi(a) \bullet_S \phi(b).$$

Definition 1.12: Isomorphism

A ring *isomorphism* is a ring homomorphism that is invertible/a bijection. If there exists an isomorphism $R \to S$, then we say R and S are isomorphic, and write $R \cong S$.